

AMCS2034 Introduction to Data Structures and Algorithms

Tutorial 9A: Graphs

- Q1. What is a graph?
- Q2. Give any **THREE (3)** applications of a graph.
- Q3. Differentiate between a directed and undirected graph.
- Q4. Explain the terms below in relation to vertices and edges of a graph.
- (i) Adjacent
 - (ii) Incident
- Q5. What is a *Tree* in the context of graph?
- Q6. Briefly explain the following edges.
- (i) Self-loop edges
 - (ii) Parallel edges
- Q7. Differentiate between a *simple path* and a *simple cycle* in a graph.
- Q8. Describe the term *bipartite graph*.
- Q9. Explain the following terms in the context of a graph:
- (i) A directed acyclic graph (DAG).
 - (ii) A forest.
 - (iii) A spanning tree.
 - (iv) A spanning forest.
- Q10. Describe the following graph traversal algorithms.
- (i) Depth-First Search
 - (ii) Breadth-First Search